

# MAT 320 Mock Midterm Solutions

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# 1 Mock Test 1

**Problem 1.1.** Find the pointwise limit of the sequence of functions  $f_n: [0, 1] \rightarrow \mathbb{R}$  defined by

$$f_n(x) = \left(x - \frac{1}{n}\right)^2.$$

Does the sequence converge uniformly to its pointwise limit?

*Solution.* The pointwise limit of  $f_n$  is the function

$$f: [0, 1] \rightarrow \mathbb{R}, \quad f(x) = x^2.$$

This is because, for each  $x \in [0, 1]$ , we see that

$$\lim_{n \rightarrow \infty} \left(x - \frac{1}{n}\right)^2 = \lim_{n \rightarrow \infty} x^2 - \frac{2x}{n} + \frac{1}{n^2} = x^2 - 2x \lim_{n \rightarrow \infty} \frac{1}{n} + \lim_{n \rightarrow \infty} \frac{1}{n^2} = x^2.$$

Moreover, this convergence is uniform. Indeed, since  $\frac{2}{n}$  and  $\frac{1}{n^2}$  both have limit 0 as  $n$  goes to infinity, for any  $\varepsilon > 0$  we can pick  $N$  such that

$$\left|\frac{2}{n}\right| + \left|\frac{1}{n^2}\right| < \varepsilon$$

(say by picking  $N$  large enough so that each term is less than  $\varepsilon/2$ ). Thus, for any  $x$ ,  $n > N$  implies

$$\begin{aligned} |f(x) - f_n(x)| &= \left|x^2 - \left(x - \frac{1}{n}\right)^2\right| \\ &= \left|\frac{2x}{n} - \frac{1}{n^2}\right| \\ &\leq \left|\frac{2x}{n}\right| + \left|\frac{1}{n^2}\right| \\ &\leq \left|\frac{2}{n}\right| + \left|\frac{1}{n^2}\right| < \varepsilon, \end{aligned}$$

which is what it means for  $f_n$  to converge to  $f$  uniformly (notably, the threshold  $N$  is independent of  $x$ .)  $\square$

**Problem 1.2.** Show that  $\mathbb{Z} \subseteq \mathbb{R}$  is closed.

*Solution.* We show that the complement of  $\mathbb{Z}$  in  $\mathbb{R}$  is open. The complement of  $\mathbb{Z}$  can be expressed as

$$\mathbb{R} \setminus \mathbb{Z} = \bigcup_{n \in \mathbb{Z}} (n, n+1),$$

since any non-integer  $x$  is in the interval  $([x], [x] + 1)$ , where  $[r]$  is the largest integer  $k$  such that  $k \leq r$ , and obviously no integers are in any of the  $(n, n+1)$ . Then,  $\mathbb{R} \setminus \mathbb{Z}$  is the union of a collection of open intervals, so it is open, thus its complement  $\mathbb{Z}$  is a closed set.  $\square$

**Problem 1.3.** Let  $(X, d)$  be a metric space. Show that if  $X$  is connected then any continuous function  $f: X \rightarrow \mathbb{Q}$  is constant.

*Solution.* The main idea of this problem is the following claim:

**Claim 1.3.1** — If a nonempty subset of  $\mathbb{Q}$  is connected, then it contains exactly one element.

*Solution.* Suppose otherwise for the sake of contradiction; that is,  $E \subseteq \mathbb{Q}$  is connected and contains two numbers  $a < b$ . Then, we have shown in a previous homework that there exists an irrational number  $r \in (a, b)$ . Consider the sets

$$A = (-\infty, r) \cap E, \quad B = (r, \infty) \cap E.$$

$A$  and  $B$  are open in  $E$  because they are intersections of open sets in  $\mathbb{R}$  with  $E$ . They are also clearly disjoint, and they are nonempty because  $a \in A$  and  $b \in B$ . Also, their union is

$$A \cup B = E \cap ((-\infty, r) \cup (r, \infty)) = E \cap (\mathbb{R} \setminus \{r\}) = E \cap \mathbb{R} \setminus E \cap \{r\} = E$$

because  $E \subseteq \mathbb{Q}$  does not contain the irrational number  $r$ . □

Now, if  $X$  is connected, then its image  $f(X)$  is also connected and nonempty, so  $f(X)$  contains exactly one number by the claim, i.e.  $f$  is a constant function. □

**Problem 1.4.** Show that the set

$$K = \{1 - 2^{-n} : n \in \mathbb{N}\} \cup \{1\} \cup \left\{ \frac{n}{1+n} : n \in \mathbb{N} \right\}$$

is a compact subset of  $\mathbb{R}$ .

*Solution.* We can do this with just the definition of compactness. Let  $\mathcal{U}$  be an open cover of  $K$ . Then, some  $U \in \mathcal{U}$  contains 1, which means that there is an open interval  $I = (1 - \varepsilon, 1 + \varepsilon) \subseteq U$ . Since

$$\lim_{n \rightarrow \infty} 1 - 2^{-n} = \lim_{n \rightarrow \infty} \frac{n}{1+n} = 1,$$

eventually  $1 - 2^{-n}$  and  $\frac{n}{1+n}$  will be in  $I$  (i.e. they will be a distance less than  $\varepsilon$  away from 1), say when  $n > N$ . That is, all but finitely many elements of  $K$  are in  $U$ .

For  $n \leq N$ , let  $A_n \in \mathcal{U}$  be an open set containing  $1 - 2^{-n}$  and let  $B_n \in \mathcal{U}$  be an open set containing  $\frac{n}{1+n}$ . Then, we have found a finite subcover of  $\mathcal{U}$  in the form

$$K \subseteq U \cup \bigcup_{n=1}^N A_n \cup \bigcup_{n=1}^N B_n,$$

since, to reiterate,  $U$  contains 1 as well as  $1 - 2^{-n}$  and  $\frac{n}{1+n}$  for  $n > N$ . □

**Problem 1.5.** Let  $I = (a, b)$  be an interval. A function  $f: I \rightarrow \mathbb{R}$  is called **convex** if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

for all  $x, y \in I$  and  $0 \leq \lambda \leq 1$ . Show that

(a) if  $f: I \rightarrow \mathbb{R}$  and  $g: \mathbb{R} \rightarrow \mathbb{R}$  are convex, and  $g$  is increasing, then their composition  $h: I \rightarrow \mathbb{R}$ ,  $h(x) = g(f(x))$ , is also convex;

(b) the function  $f$  is convex if and only if

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(y)}{z - y}$$

for all  $x < y < z$  in  $I$ ;

(c) if  $f$  is of class  $C^2$  and  $f''(x) \geq 0$  at every  $x \in I$  then  $f$  is convex.

*Solution.* For (a), let  $\lambda \in [0, 1]$  and  $x, y \in I$ . Then

$$\begin{aligned} h(\lambda x + (1 - \lambda)y) &= g(f(\lambda x + (1 - \lambda)y)) \\ &\leq g(\lambda f(x) + (1 - \lambda)f(y)) \\ &\leq \lambda g(f(x)) + (1 - \lambda)g(f(y)) \\ &= \lambda h(x) + (1 - \lambda)h(y), \end{aligned}$$

with the second line using the fact that  $f$  is convex and  $g$  is increasing in the form

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &\leq \lambda f(x) + (1 - \lambda)f(y) \\ \implies g(f(\lambda x + (1 - \lambda)y)) &\leq g(\lambda f(x) + (1 - \lambda)f(y)) \end{aligned}$$

and the third line using the fact that  $g$  is convex.

To prove (b), assume first that  $f$  is convex. Then, suppose  $x < y < z$  are elements of  $I$ . We can write

$$\lambda = \frac{z - y}{z - x}$$

(which is in  $(0, 1)$  because  $0 < z - y < z - x < 1$ ) so that  $\lambda x + (1 - \lambda)z = y$ . Thus, convexity says that

$$f(y) \leq \lambda f(x) + (1 - \lambda)f(z) = \frac{z - y}{z - x} \cdot f(x) + \frac{y - x}{z - x} \cdot f(z).$$

This can be rearranged to

$$\frac{z - y}{z - x}(f(y) - f(x)) \leq \frac{y - x}{z - x}(f(z) - f(y)),$$

which is equivalent to

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(y)}{z - y}$$

after multiplying both sides by  $\frac{z - x}{(z - y)(y - x)}$ .

To prove the converse, we can reverse the algebraic manipulation to get

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(y)}{z - y} \implies f(y) \leq \frac{z - y}{z - x} \cdot f(x) + \frac{y - x}{z - x} \cdot f(z)$$

whenever  $x < y < z$ . For any  $x < z$  in  $I$  and  $\lambda \in [0, 1]$ , we can then let  $y = \lambda x + (1 - \lambda)z$ , which is between  $x$  and  $z$ . Solving for  $\lambda$  lets us recover

$$\lambda = \frac{z - y}{z - x},$$

so the given inequality implies that

$$f(\lambda x + (1 - \lambda)y) = f(y) \leq \frac{z - y}{z - x} \cdot f(x) + \frac{y - x}{z - x} \cdot f(z) = \lambda f(x) + (1 - \lambda)f(z).$$

Finally, for (c), let  $x < y < z$  be in  $I$ . Since  $f$  is class  $C^1$ , the mean value theorem says that there exist  $x < \alpha < y$  and  $y < \beta < z$  such that

$$\frac{f(y) - f(x)}{y - x} = f'(\alpha), \quad \frac{f(z) - f(y)}{z - y} = f'(\beta).$$

Since  $f'' \geq 0$ , the function  $f'$  is increasing, so  $\alpha < y < \beta$  implies  $f'(\alpha) \leq f'(\beta)$ . Thus,

$$\frac{f(y) - f(x)}{y - x} = f'(\alpha) \leq f'(\beta) = \frac{f(z) - f(y)}{z - y}$$

for any  $x < y < z$ , which by part (b) implies that  $f$  is convex. □

**Problem 1.6.** We construct an infinite sequence  $\mathbb{R} \supset [0, 1] \supset S_1 \supset S_2 \supset \dots$  inductively. Let

$$S_1 = [0, 1/3] \cup [2/3, 1]$$

be obtained from  $[0, 1]$  by removing the middle third open interval  $(1/3, 2/3)$ . Repeat the process, removing the middle third open intervals from  $[0, 1/3]$  and  $[2/3, 1]$  to obtain

$$S_2 = [0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1].$$

In general,  $S_{n+1}$  is obtained from removing the middle third of each interval in  $S_n$ . Define

$$C = \bigcap_{n=1}^{\infty} S_n,$$

which is known as the **Cantor set**. Prove that

- (a)  $C$  is compact.
- (b) there is no nonempty set  $A \subseteq C$  that is open in  $\mathbb{R}$ .
- (c)  $C$  has infinitely many points.

*Solution.* The main fact that we will use about the structure of  $S_n$  is:

**Claim 1.6.1** — The set  $S_n$  can be expressed as the union of  $2^n$  pairwise disjoint closed intervals, each with length  $3^{-n}$  (where “length” is the absolute difference between the endpoints).

*Proof.* This is a simple induction proof. For  $S_1$ , we can verify manually that it is a union of  $2^1$  disjoint intervals  $[0, 1/3]$  and  $[2/3, 1]$  and that both intervals have length  $3^{-1}$ .

Now, if the claim holds for  $S_n$ , then write

$$S_n = \bigcup_{k=1}^{2^n} [a_k, a_k + 3^{-n}]$$

where the  $2^n$  intervals  $[a_k, a_k + 3^{-n}]$  are pairwise disjoint. Then, after removing the middle third of each interval, we get

$$S_{n+1} = \bigcup_{k=1}^{2^n} [a_k, a_k + 3^{-n-1}] \cup [a_k + 2 \cdot 3^{-n-1}, a_k],$$

and each of these  $2 \cdot 2^n$  intervals has length  $3^{-n-1}$  (it is easy to check that pairwise disjointness is preserved).  $\square$

Now, all the parts fall pretty quickly.

- (a) For (a), we can see that  $C$  is bounded because it is a subset of the bounded set  $[0, 1]$ , and it is closed because it is the intersection of the  $S_n$ , which are each closed because they are unions of finitely many ( $2^n$ ) closed intervals. Thus,  $C$  is a closed and bounded subset of  $\mathbb{R}$ , so it is compact.



(b) For (b), suppose for the sake of contradiction that  $A \subseteq C$  is a nonempty open subset of  $\mathbb{R}$ . Then, let  $x \in A$ . There exists  $\varepsilon > 0$  such that  $(x - \varepsilon, x + \varepsilon) \subseteq C$ , meaning that it is a subset of all the  $S_n$ . Pick  $N$  so that  $3^{-N} < 2\varepsilon$ . By the claim, we can write  $S_N$  as a union of pairwise disjoint open intervals of length  $3^{-N}$ , one of which, say  $I = [a, a + 3^{-N}]$ , contains  $x$ . Then, we can find a  $\delta$  such that  $(a - \delta, a)$  contains no points of  $C$ , as otherwise, we could find two disjoint closed intervals whose union is a closed interval, which is impossible because all intervals are connected. To finish,  $(x - \varepsilon, x + \varepsilon)$  contains some points in  $(a - \delta, a)$ , so it is not a subset of  $C$ , hence  $A$  cannot be a subset of  $C$ .

(c) Observe that if

$$S_n = \bigcup_{k=1}^{2^n} [a_k, a_k + 3^{-n}],$$

then each of the  $a_k$  is also in  $S_{n+1}$  because the endpoints of the intervals are obviously not in the middle third of the respective interval. Thus, each  $S_n$  contains at least  $2^n$  points, so  $C$  contains at least  $2^n$  points for each  $n$ , meaning it contains infinitely many points.

□

## 2 Mock Test 2

**Problem 2.1.** Find the radius of convergence of the power series

$$f(x) = \sum_{n=1}^{\infty} \frac{n^3}{3^n} x^n.$$

*Solution.* The radius of convergence of  $f$  is equal to

$$\left( \limsup \left( \frac{n^3}{3^n} \right)^{1/n} \right)^{-1} = \left( \limsup \frac{n^{3/n}}{3} \right)^{-1} = \left( \lim_{n \rightarrow \infty} \frac{(n^{1/n})^3}{3} \right)^{-1} = \left( \frac{1^3}{3} \right)^{-1} = \boxed{3}.$$

□

**Problem 2.2.** Let  $(X, d)$  be a metric space and  $f: X \rightarrow \mathbb{R}$  a continuous function. Show that the set of points

$$P = \{x \in X : f(x) > 0\}$$

where the function  $f$  is positive is an open subset of  $X$ .

*Proof.* The set of positive reals

$$\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\} = (0, \infty)$$

is open in  $\mathbb{R}$ , so continuity of  $f$  shows that  $f^{-1}(\mathbb{R}^+) = P$  is open in  $X$ .  $\square$

**Problem 2.3.** Show that the set

$$K = \left\{ x \in \mathbb{R} : \left| x - \frac{1}{n} \right| \leq \frac{1}{3n^2 + n} \text{ for some } n \in \mathbb{N} \right\} \cup \{0\}$$

is compact.

*Solution.* We can rewrite  $K$  as the union

$$K = \{0\} \cup \bigcup_{n=1}^{\infty} \left[ \frac{1}{n} - \frac{1}{3n^2 + n}, \frac{1}{n} + \frac{1}{3n^2 + n} \right].$$

Although this is a union of infinitely many open intervals, we are lucky in that they are pairwise disjoint because

$$\begin{aligned} \frac{1}{n+1} + \frac{1}{3(n+1)^2 + (n+1)} &< \frac{1}{n+1} + \frac{1}{3n^2 + n} \\ &= \frac{1}{n} - \frac{1}{n^2 + n} + \frac{1}{3n^2 + n} \\ &\leq \frac{1}{n} - \frac{2}{3n^2 + n} + \frac{1}{3n^2 + n} \\ &= \frac{1}{n} - \frac{1}{3n^2 + n} \end{aligned}$$

for all positive integers  $n$ ; the second-to-last line follows from

$$\frac{1}{n^2 + n} > \frac{2}{3n^2 + n} \iff 3n + 1 \geq 2n + 2.$$

Also, we can easily see that each interval only contains positive numbers. We can thus write  $K$  as the complement of the open set

$$(-\infty, 0) \cup \left( \frac{5}{4}, \infty \right) \cup \bigcup_{n=1}^{\infty} \left( \frac{1}{n+1} + \frac{1}{3(n+1)^2 + (n+1)}, \frac{1}{n} - \frac{1}{3n^2 + n} \right),$$

so  $K$  is closed.  $K$  is also clearly bounded as it is a subset of  $[0, 5/4]$ , so it is closed and bounded, hence compact.  $\square$

**Problem 2.4.** *Prove that the equation  $xe^x = 2$  has a solution  $x \in (0, 1)$ .*

*Solution.* We showed in class that

$$f(x) = xe^x = \sum_{k=0}^{\infty} \frac{x^{k+1}}{k!}$$

is continuous on its radius of convergence (which in this case is  $\mathbb{R}$ ). Note also that

$$f(0) = 0 < 2 < 1 + 1 + \sum_{k=2}^{\infty} \frac{1}{k!} = \sum_{k=0}^{\infty} \frac{1}{k!} = f(1),$$

so, by the intermediate value theorem, there is an  $r \in (0, 1)$  such that  $2 = f(r) = re^r$ .  $\square$

**Problem 2.5.** We say that a function  $u: [0, 1] \rightarrow \mathbb{R}$  is a **Lipschitz function** if there is a constant  $L > 0$  such that

$$|u(x) - u(y)| \leq L|x - y|$$

for all  $x, y \in [0, 1]$ . For a function  $u: [0, 1] \rightarrow \mathbb{R}$  define

$$[u] = \sup \left\{ \frac{|u(x) - u(y)|}{|x - y|} : x, y \in [0, 1] \text{ with } x \neq y \right\},$$

and

$$\|u\| = \sup\{|u(x)| : x \in [0, 1]\}.$$

Let  $X = \{u: [0, 1] \rightarrow \mathbb{R} : [u] < \infty\}$ . Perform the following tasks:

(a) Show that a function  $u: [0, 1] \rightarrow \mathbb{R}$  is Lipschitz if and only if  $[u] < \infty$ , and explain why all class  $C^1$  functions belong to  $X$ .

(b) Show that

$$d'(u, v) = [u - v], \quad u, v \in X$$

satisfies the triangle inequality. Is  $(X, d')$  a metric space?

(c) Show that  $\|u\| + \|v\| < \infty$  for all Lipschitz functions and that

$$d(u, v) = [u - v] + \|u - v\|$$

is a distance function on  $X$ .

(d) Show that  $(X, d)$  is a complete metric space.

*Solution.* For (a), observe that if  $u$  is Lipschitz then there exists an  $L$  such that

$$|u(x) - u(y)| \leq L|x - y|$$

for all  $x, y \in [0, 1]$ . This always holds trivially when  $x = y$ , so it suffices to loosen this to all  $x \neq y$  in  $[0, 1]$ . That is, we can safely say that  $u$  is Lipschitz iff

$$\frac{|u(x) - u(y)|}{|x - y|} \leq L$$

for all  $x \neq y$  in  $[0, 1]$ ; that is, if  $u$  is Lipschitz, the supremum of the difference quotient (which is  $[u]$ ) over all pairs of distinct reals in  $(0, 1)$  is at most  $L$ , i.e. less than  $\infty$ . Conversely, if the supremum is a real number, say  $L$ , then

$$\frac{|u(x) - u(y)|}{|x - y|} \leq L$$

for all  $x, y \in [0, 1]$  with  $x \neq y$  by definition of supremum.

Additionally, if  $u$  is class  $C^1$ , then its derivative is always at most  $L$  (because the difference quotient is at most  $L$ ), so  $u'$  is bounded. Then, for any  $x < y$ , we can find  $\xi$  between  $x$  and  $y$  such that

$$\frac{u(x) - u(y)}{x - y} = u'(\xi),$$

which means that the difference quotient is bounded as long as  $u'$  is bounded, and we established above that  $u'$  is bounded, thus  $u \in X$ .

For (b), we just observe that, for each  $x \neq y$ ,

$$\frac{|(u-v)(x) - (u-v)(y)|}{|x-y|} + \frac{|(v-w)(x) - (v-w)(y)|}{|x-y|} \geq \frac{|(u-w)(x) - (u-w)(y)|}{|x-y|}$$

by the triangle inequality, so

$$\begin{aligned} [u-v] + [v-w] &\geq \sup_{0 \leq x < y \leq 1} \frac{|(u-v)(x) - (u-v)(y)|}{|x-y|} + \frac{|(v-w)(x) - (v-w)(y)|}{|x-y|} \\ &\geq \sup_{0 \leq x < y \leq 1} \frac{|(u-w)(x) - (u-w)(y)|}{|x-y|} \\ &= [u-w]. \end{aligned}$$

However,  $d'$  is not a metric on  $X$ . For example, consider the functions  $u(x) = 1$  and  $v(x) = 2$ , both of which are in  $X$  (because they're class  $C^1$ ). We get

$$[u-v] = \sup_{0 \leq x < y \leq 1} \left| \frac{(2-1) - (2-1)}{x-y} \right| = 0,$$

even though  $u \neq v$ .

For (c), suppose that  $u$  is Lipschitz. Then,  $[u]$  is a real number by part (a), and  $\|u\|$  is a real number because  $u$  being Lipschitz means that there exists an  $L$  such that

$$|u(x)| - |u(0)| \leq |u(x) - u(0)| \leq L \cdot |x - 0| \leq L \implies |u(x)| \leq L + |u(0)|$$

for all  $x$ . Now we check that  $d$  is a metric on  $X$ .

- We can check easily that  $\|u\|$  satisfies the triangle inequality (in much the same way as the previous part), so the sum  $[u] + \|u\|$  also satisfies the triangle inequality.
- Moreover, it is trivial to check that  $d(u, v) = d(v, u)$ , as the absolute values within the suprema mean that  $[u-v] + \|u-v\| = [v-u] + \|v-u\|$ .
- Since  $[u]$  and  $\|u\|$  are always nonnegative, we only need to check when  $[u] + \|u\| = 0$ . We can see that  $d(u, v) = 0$  implies that  $\|u-v\| = 0$ , which then forces  $|(u-v)(x)| \leq 0$  for all  $x$ , i.e.  $u-v \equiv 0 \iff u=v$ . It is trivial to check that  $u=v$  implies  $d(u, v) = 0$ , so  $d$  is positive definite.

Finally, for (d), we need to show that  $(X, d)$  is a complete metric space. Let  $(u_n)$  be a Cauchy sequence in  $X$ ; we need to show that it converges in  $X$ . The sequence being Cauchy means that, for each  $\varepsilon > 0$ , there exists  $N$  such that  $m, n > N$  implies  $d(u_m, u_n) < \varepsilon$ . For such  $m$  and  $n$ , we can bound

$$\varepsilon > d(u_m, u_n) = [u_m - u_n] + \|u_m - u_n\| \geq |u_m(x) - u_n(x)|,$$

so in fact for any fixed  $x$  the sequence  $(u_n(x))$  is a Cauchy sequence of real numbers; thus  $(u_n)$  has a pointwise limit  $u$ . In fact, this convergence is uniform because  $(u_m)$  being Cauchy with respect to  $d$  means that it is Cauchy with respect to the standard

$$\|u - v\| = \sup_{x \in [0,1]} |u(x) - v(x)|$$

metric on bounded functions.

I claim that  $u$  is the limit of  $(u_n)$  in  $(X, d)$ . First, we show that  $u \in X$ . For  $m, n > N$ , we can say that

$$\varepsilon > d(u_m, u_n) = [u_m - u_n] + \|u_m - u_n\| \geq [u_m - u_n] \geq |[u_m] - [u_n]|$$

by triangle inequality. That is,  $([u_n])$  is a Cauchy sequence, so in particular the constants  $L_n$  such that

$$|u_n(x) - u_n(y)| \leq L_n \cdot |x - y|$$

have a limit  $L$ . Thus, we can see that

$$|u(x) - u(y)| \leq |u(x) - u_n(x)| + |u_n(x) - u_n(y)| + |u_n(y) - u(y)|,$$

so the left-hand side is less than  $L|x - y|$  by taking  $n \rightarrow \infty$ , i.e.  $u$  is Lipschitz, thus in  $X$ . Now, for conciseness, define

$$[v]_{x,y} = \frac{|v(x) - v(y)|}{|x - y|}$$

where  $x \neq y$  and  $v \in X$ . We wish to bound

$$d(u, u_n) = [u - u_n] + \|u - u_n\|.$$

Since  $u_n \rightarrow u$  uniformly, we can bound the second term by  $\varepsilon/2$ . For the first term, we bound pointwise: take any  $x \neq y$ , and then observe

$$[u - u_n]_{x,y} \leq [u - u_m]_{x,y} + [u_m - u_n]_{x,y}.$$

For  $m, n > M$ , the second addend on the RHS can be bounded by say  $\frac{\varepsilon}{4}$ , and, for  $m > M'$ , the first addend can be bounded by  $\frac{\varepsilon}{4}$  because  $u_m \rightarrow u$  pointwise. More explicitly,

$$\frac{|u(x) - u_m(x) - u(y) + u_m(y)|}{|x - y|} \leq \frac{|u(x) - u_m(x)| + |u(y) - u_m(y)|}{|x - y|},$$

and we can bound the numerator terms by any positive real as long as we make  $n$  large enough, so we can bound both by  $\frac{\varepsilon|x-y|}{8}$ . Thus,

$$[u - u_n]_{x,y} \leq \frac{\varepsilon}{2}$$

for all  $x \neq y$  in  $[0, 1]$ , and now  $\frac{\varepsilon}{2}$  is a constant bound so we can just take the supremum, which gives us  $[u - u_n] \leq \frac{\varepsilon}{2}$ , which finishes.  $\square$



**Problem 2.6.** Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a differentiable function such that

$$\lim_{x \rightarrow \infty} f(x) = 10, \quad \lim_{x \rightarrow -\infty} f(x) = -10.$$

Let  $Z = \{z \in \mathbb{R} : f(z) = 0\}$  denote the zero set of  $f$ . We say that  $z \in Z$  is an **isolated zero** if there is an open set  $U$  containing  $z$  such that  $f(x) \neq 0$  for all  $x \in U \setminus \{z\}$ . Perform the following tasks.

- (a) Show that  $Z$  is a compact set.
- (b) Assuming that  $f$  only has isolated zeroes show that  $Z$  is finite.
- (c) Assuming there is no point  $z \in \mathbb{R}$  such that  $f(z) = f'(z) = 0$  show that  $Z$  is finite.

*Solution.* To prove (a), we prove that  $Z$  is closed and bounded. Since  $f$  is differentiable, it is continuous, so the pre-image  $Z = f^{-1}(\{0\})$  of the closed set  $\{0\}$  is also closed. To see that it is bounded, we see that the limit conditions say that for each  $\varepsilon > 0$ , there exist  $N_1 > N_2$  such that

$$x > N_1 \implies |f(x) - 10| < \varepsilon, \quad x < N_2 \implies |f(x) + 10| < \varepsilon.$$

Picking  $\varepsilon = 1$ , this means that  $f(x)$  is nonzero when  $x > N_1$  or  $x < N_2$ ; thus,  $Z \subseteq [N_2, N_1]$ , so it is bounded.

For (b), the definition of isolated zeroes means that for each  $r \in Z$ , we can find a  $U_r$  such that  $f(x) \neq 0$  for all  $x \in U_r \setminus \{r\}$ . That is,  $U_r \cap Z = \{r\}$  for all  $r \in Z$ . Now, consider the open cover  $\mathcal{U} = \{U_r : r \in Z\}$ . Any proper subset of this open cover no longer covers  $Z$ , since for each  $r \in Z$  there is exactly one set in  $\mathcal{U}$  that contains  $r$ ; thus, the only subcover of  $\mathcal{U}$  is  $\mathcal{U}$ . On the other hand,  $\mathcal{U}$  has a finite subcover because  $Z$  is compact, so  $\mathcal{U}$  is finite, hence so is  $Z$  (because we have a bijection  $\mathcal{U} \rightarrow Z$  given by  $U_r \mapsto r$ ).

Finally, for (c), we show that  $f$  only has isolated zeroes. Let  $z \in Z$ . Then,

$$\lim_{x \rightarrow z} \frac{f(z) - f(x)}{z - x} = f'(z)$$

is nonzero, so the difference quotient is nonzero within some deleted neighborhood  $(z - \varepsilon, z + \varepsilon) \setminus \{z\}$  of  $z$ . But then

$$\frac{f(z) - f(x)}{z - x} = \frac{-f(x)}{z - x} \neq 0$$

whenever  $0 < |x - z| < \varepsilon$ , which means that we have found an open set  $U = (z - \varepsilon, z + \varepsilon)$  such that  $f(x) \neq 0$  whenever  $x \in U \setminus \{z\}$ , hence  $z$  is an isolated zero. We can then conclude by (b) that  $Z$  is finite!  $\square$